A Qualitative Strong Open Set Condition Does Not Force Uniform Prefractal Domain Constants

Automated Math Discovery Candidate Counterexample

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Status. Candidate full counterexample to Conjecture 1 of arXiv:2003.10134 under Assumptions 1 and 3 as printed, likely valid pending expert review. A single pair of fixed convex open polygons satisfies the strong open set condition at every prefractal level, but the best Jones uniformity constants tend to zero.

Source Conjecture

Dekkers, Rozanova-Pierrat, and Teplyaev consider polygonal faces $K_m = \Phi_m(K_0)$, generated by finite level-dependent families of contracting similitudes, and the associated domains Ω_m. Their Assumption 1 requires each K_m to have the same codimension-two boundary as K_0. Assumption 3 requires two fixed nested convex open polygons to satisfy the open set condition for every Φ_m. Conjecture 1 then says that Ω_m and the limit Ω are uniformly exterior and interior (ε, ∞)-domains with ε independent of m [1].

where $B_r(x) \subset \mathbb{R}^2$ denotes the Euclidean ball of radius r and centered at x with

$$d^{(\xi)} = \frac{\ln 4}{p_1 \ln p_1 + p_2 \ln p_2}.$$

According to Definition 3, it means that $K^{(\xi)}$ is a $d^{(\xi)}$-set and the measure $\mu^{(\xi)}$ is a $d^{(\xi)}-$dimensional measure equivalent to the $d^{(\xi)}$-dimensional Hausdorff measure $m_{d^{(\xi)}}$.

We now discuss the more general set-up in Subsection 6.1. The standard Open Set Condition [26, Section 9.2] is satisfied for an iterated function system if there is a non-empty bounded open set O such that $\Phi_m(O) \subset O$ with the union in the left-hand side disjoint. Note that the open set O may not be unique. Conjecture 1 assumes The Fractal Self-Similar Face Condition (Assumption 1) and a strong version of the Open Set Condition, see Figure 2, that we introduce as follows.

Assumption 3 (A Strong Open Set Condition)**.** *We assume the Open Set Condition for the sequence Φ_m is satisfied with two different convex open polygons $\mathcal{O} \subsetneq \mathcal{O}'$, not depending on m, such that*

$$\partial\mathcal{O} \cap K_0 = \partial\mathcal{O}' \cap K_0 = \partial\mathcal{O} \cap \partial\mathcal{O}' = \partial_{(n-2)} K_0.$$

Conjecture 1. *:* If Assumptions 1 and 3 are satisfied, then Ω_m and Ω are uniformly exterior and interior (ϵ, ∞)-domains, that is, ϵ does not depend on m.

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We disprove the stated implication. The issue is that disjointness of open cells is qualitative: it supplies no lower bound on a narrowing passage.

Construction

Work in $\mathbb{R}^2$. Put

$$A = (-100, 0), \qquad B = (100, 0), \qquad K_0 = [A, B].$$

For $h > 0$, let

$$O_h = \operatorname{int}\operatorname{conv}\{A, (0, h), B, (0, -h)\}.$$

We use the two fixed convex polygons

$$O = O_{3/2} \subsetneqq O' = O_2.$$

Their boundaries satisfy

$$\partial O \cap K_0 = \partial O' \cap K_0 = \partial O \cap \partial O' = \{A, B\} = \partial_{(0)} K_0,$$

exactly as required by Assumption 3.

Fix $m \geq 2$, and set

$$a_m = \frac{1}{m}, \qquad h_m = \frac{1}{m^4}, \qquad J_m = (m-1)m^3.$$

The levels $t_j = a_m + jh_m$, $0 \leq j \leq J_m$, run from a_m to 1. Let C_m be the polygonal arc traversing, in order,

$$A, \quad (-t_j^2, -1 + t_j) \quad (j = J_m, J_m - 1, \ldots, 0),$$
$$(a_m^2, -1 + a_m), \quad (t_j^2, -1 + t_j) \quad (j = 1, 2, \ldots, J_m), \quad B.$$

Thus C_m follows two finely polygonalized parabolic arms and closes their lower ends with a horizontal cap of width $2a_m^2$.

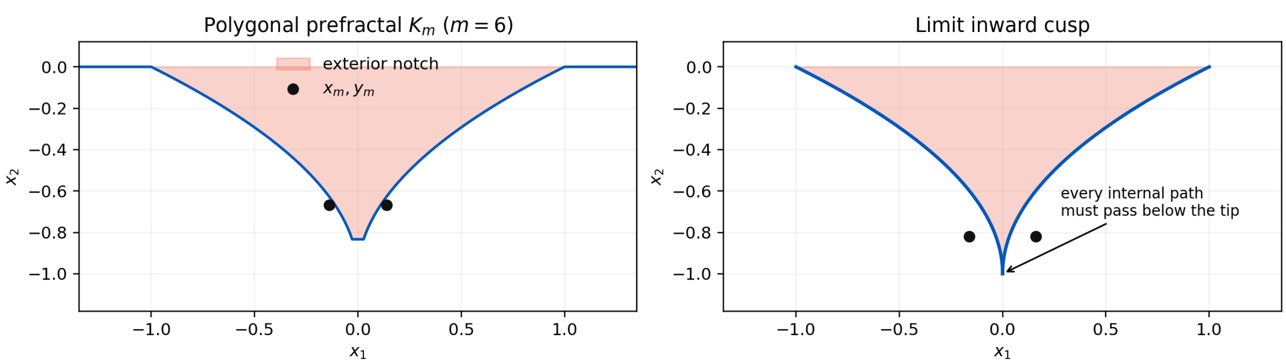
For every oriented edge $e = [p, q]$ of C_m, let $\psi_{e,m}$ be the orientation-preserving similitude sending A to p and B to q, and put

$$\Phi_m(E) = \bigcup_{e \subset C_m} \psi_{e,m}(E).$$

Then $C_m = \Phi_m(K_0)$. Complete C_m by the three fixed lower sides

$$B \longrightarrow (100, -2) \longrightarrow (-100, -2) \longrightarrow A$$

and let Ω_m be the bounded Jordan domain below C_m. Equivalently, it is the rectangle $(-100, 100) \times (-2, 0)$ with the capped cusp-shaped notch above C_m removed.



The Strong OSC Still Holds

Lemma 1. *For every $m \geq 2$, the family $\{\psi_{e,m}\}_e$ consists of contractions and satisfies the open set condition with both fixed sets O and O'. Moreover C_m has zero-dimensional boundary $\{A, B\}$.*

Proof. The two baseline edges have length 99, every arm edge has length at most $\sqrt{5}h_m < 3m^{-4}$, and the cap has length $2m^{-2}$. All are shorter than $|A - B| = 200$, so the similitudes are contractions.

The image of O_h under the map associated with an edge of length L is the open rhombus having that edge as its long diagonal and transverse half-diagonal $hL/200$. On a baseline edge these are homothetic subrhombi of O_h. Every other edge lies in the central box $[-1, 1] \times [-1, 0]$, while its cell is separated from ∂O_h by a fixed margin. Hence

$$\Phi_m(O_h) \subset O_h, qquad h \in \{3/2, 2\}.$$

It remains to check disjointness. Consecutive cell rhombi have disjoint open interiors and share only their common long-diagonal endpoint: their aperture is at most $\arctan(1/50)$, whereas the outward angle between consecutive edges is bounded away from twice this aperture. On either parabolic arm, two nonconsecutive cells have a full intervening vertical increment h_m, while each transverse vertical displacement is at most $h_m/50$. Cells on opposite arms are horizontally separated by at least $2m^{-2}$, whereas the sum of their transverse widths is at most $m^{-4}/50$. The cap cell meets only the first arm cell on each side; all other cap–arm pairs are separated by the same cone or coordinate estimates. The baseline cells are separated from all nonadjacent cusp cells. Thus the open images $\psi_{e,m}(O')$ are pairwise disjoint. Since $\psi_{e,m}(O) \subset \psi_{e,m}(O')$, the same is true for O.

Finally, C_m is a simple polygonal arc from A to B, so its zero-dimensional hypersurface boundary is exactly $\{A, B\}$. This proves both Assumptions 1 and 3. (The retained verifier checks all cell pairs, containments, and contraction ratios independently at representative levels.) $\square$

Failure of Every Uniform Constant

Theorem 1. *The domains Ω_m satisfy Assumptions 1 and 3 of the source, but there is no $\varepsilon > 0$ for which they are all interior (ε, ∞)-domains. In fact, if ε_m is any admissible constant for Ω_m, then*

$$\varepsilon_m \leq \frac{5}{m}.$$

Their limit Ω is not a uniform domain either. Consequently Conjecture 1 is false under its printed hypotheses.

Proof. For $m \geq 3$, the level $t = 2a_m$ occurs in the mesh because $(2a_m - a_m)/h_m = m^3$. At height $-1 + 2a_m$, the two arms of C_m are at first coordinates $\pm 4a_m^2$. Therefore

$$x_m = (-5a_m^2, -1 + 2a_m), \qquad y_m = (5a_m^2, -1 + 2a_m)$$

belong to Ω_m, and

$$|x_m - y_m| = 10a_m^2.$$

Let $\gamma \subset \Omega_m$ be any rectifiable arc joining these points. Continuity of its first coordinate gives a point $z \in \gamma$ with $z_1 = 0$. The notch contains the entire central segment on the line $x_1 = 0$ above its cap, so necessarily $z_2 < -1 + a_m$. Hence each part of γ from an endpoint to z has length at least a_m, and

$$\ell(\gamma) \geq 2a_m.$$

Condition (i) in the source definition of an (ε, ∞)-domain would require

$$2a_m \leq \ell(\gamma) \leq \frac{|x_m - y_m|}{\varepsilon_m} = \frac{10a_m^2}{\varepsilon_m}.$$

Thus $\varepsilon_m \leq 5a_m = 5/m$, which tends to zero. The interior conclusion of Conjecture 1 already fails, regardless of the exterior claim.

The polygonal arms converge uniformly to

$$x_1 = \pm t^2, qquad x_2 = -1 + t, qquad 0 \leq t \leq 1,$$

and the caps shrink to the cusp tip. Thus Ω_m converge both in Hausdorff-boundary distance and in characteristic functions to the rectangle with this inward parabolic notch. For the limit, use $x_t = (-5t^2, -1 + 2t)$ and $y_t = (5t^2, -1 + 2t)$, or rescale the preceding argument. Their distance is $10t^2$, while every internal joining arc has length at least $2t$; letting $t \downarrow 0$ proves that Ω is not uniform. $\square$

Interpretation and Review Focus

The construction uses precisely the level-dependent notation $(\psi_{i,m})_{1 \leq i \leq N_m}$ in Subsection 6.1. If the appendix's successive-composition convention is intended instead, apply the m-th cusp generator within every cell produced at earlier levels. Similarity invariance leaves a copied obstruction $\varepsilon \leq 5/m$, and the fixed open-set conditions compose. Thus the defect is not an artifact of placing the generator only at the top scale.

A stationary finite generator library, a uniform lower contraction ratio, or a quantitative relative-separation condition could plausibly repair the claim. None appears in Assumptions 1 and 3. Expert review should therefore focus on whether an unstated bounded-geometry convention was intended; under the literal assumptions, the qualitative OSC cannot control the Harnack-chain constant across levels.

A bounded novelty search through August 11, 2026 checked the run indexes, exact conjecture wording, the published DOI, and the eleven citing works listed by OpenAlex. No later primary source resolving Conjecture 1 was found.

References

[1] A. Dekkers, A. Rozanova-Pierrat, and A. Teplyaev, *Mixed boundary valued problems for linear and nonlinear wave equations in domains with fractal boundaries*, Calc. Var. Partial Differential Equations 61 (2022), article 75; arXiv:2003.10134v4, Assumption 3 and Conjecture 1, p. 35, DOI: 10.1007/s00526-021-02159-3.

[2] P. W. Jones, *Quasiconformal mappings and extendability of functions in Sobolev spaces*, Acta Math. 147 (1981), 71–88.